CHAPTER 7

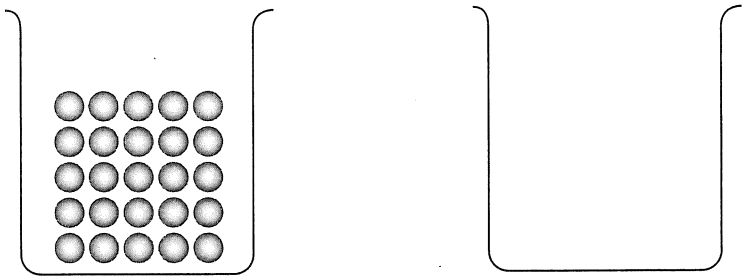
Physical and chemical change

7.1 Knowledge preview

Science understanding

FOUNDATION	STANDARD	ADVANCED

1 Look carefully at the diagram below.



(a) Use the empty beaker to re-draw the particles of the melted solid after heat has been applied to the beaker.

(b) Is the change that you showed a physical change or a chemical change? Explain your answer.

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2 Identify each of the following as a physical change (P) or a chemical change (C) by writing P or C in the space provided.

(a) Ice melts to water. _____

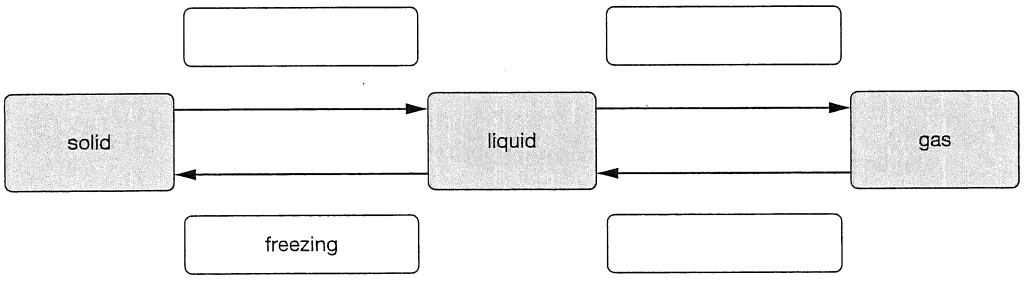
(b) Glucose breaks down to carbon dioxide and water during respiration. _____

(c) An aluminium can is squashed to take up less space in the rubbish bin. _____

(d) Sodium bicarbonate releases carbon dioxide to make the cake rise. _____

(e) An egg is cooked. _____

3 Complete the flow chart using the terms: melting, freezing, condensation, evaporation. The first one has been done for you.



7.6 Reverse processes

Science understanding

FOUNDATION | STANDARD | ADVANCED

1 For each physical change in the left column, identify the name of its reverse change in the right column and draw a line between the two. Refer to Figure 7.6.1 to help you.

Physical change
(a) condensing
(b) freezing
(c) separating
(d) crystallising
(e) expansion
(f) evaporating
(g) mixing
(h) contraction
(i) melting
(j) dissolving

Reverse change
freezing
expansion
separating
evaporating
dissolving
mixing
contraction
crystallising
melting
condensing

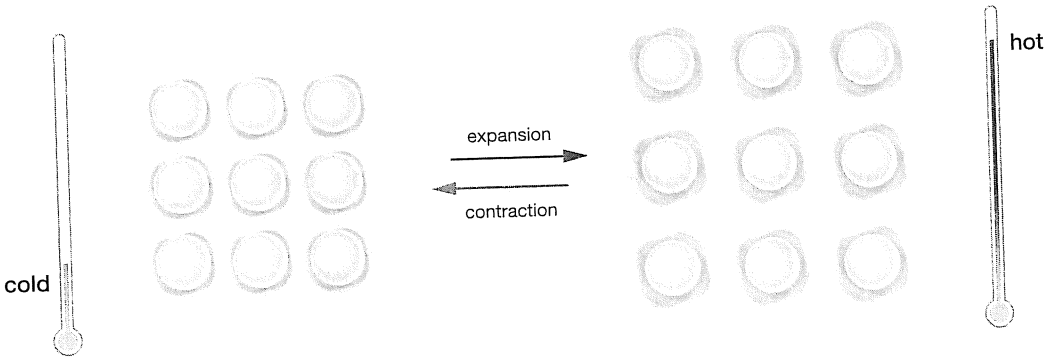


Figure 7.6.1 Expansion and contraction changes in solids and liquids

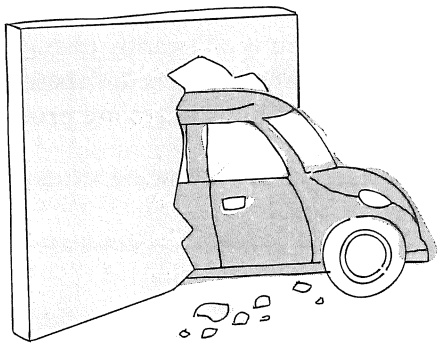
RATE MY UNDERSTANDING
Shade the face that shows your rating

(g) stirring a cake mixture



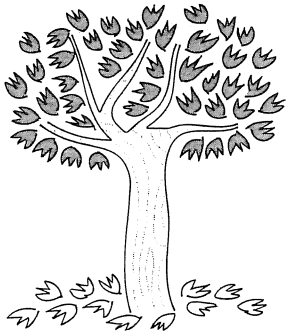
physical ☐ chemical ☐

(h) crashing a car



physical ☐ chemical ☐

(i) autumn leaves changing colour



physical ☐ chemical ☐

(j) an explosion



physical ☐ chemical ☐

(k) making a new chemical



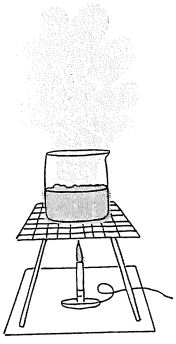
physical ☐ chemical ☐

(l) baking bread



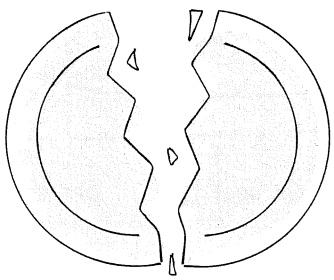
physical ☐ chemical ☐

(m) boiling water



physical ☐ chemical ☐

(n) breaking a plate



physical ☐ chemical ☐

RATE MY UNDERSTANDING
Shade the face that shows your rating ☺ 😐 ☹

7.2 Physical and chemical change

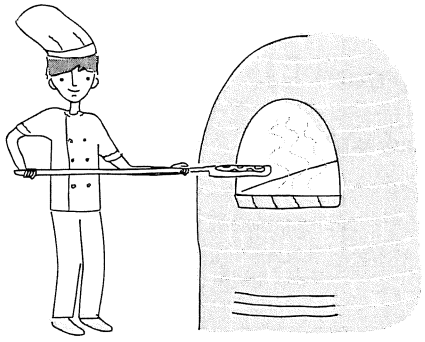
Science understanding

FOUNDATION	STANDARD	ADVANCED

The world around you is constantly changing. Some of these changes are known as physical changes, in which there are no new substances produced. Other changes are known as chemical changes, in which new substances are produced.

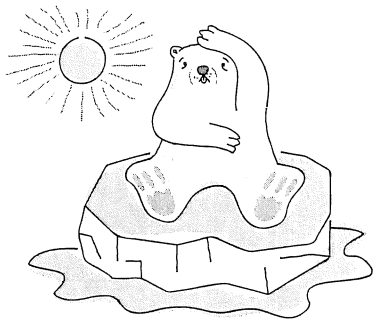
1 Classify each of the following situations as either a physical or chemical change by ticking the correct box.

(a) cooking a pizza



physical ☐ chemical ☐

(b) ice melting



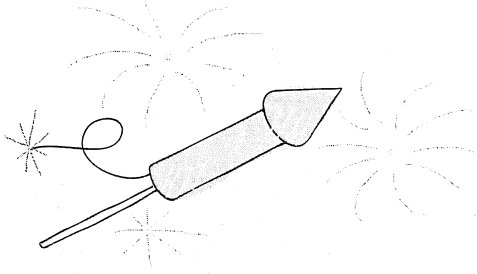
physical ☐ chemical ☐

(c) chopping wood



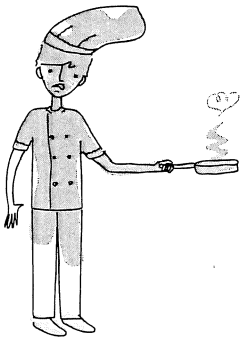
physical ☐ chemical ☐

(d) fireworks exploding



physical ☐ chemical ☐

(e) burning breakfast



physical ☐ chemical ☐

(f) ice-cream hitting the ground



physical ☐ chemical ☐

Year 8 Science

Home reading and Homework

WEEK EIGHT

Name: _____

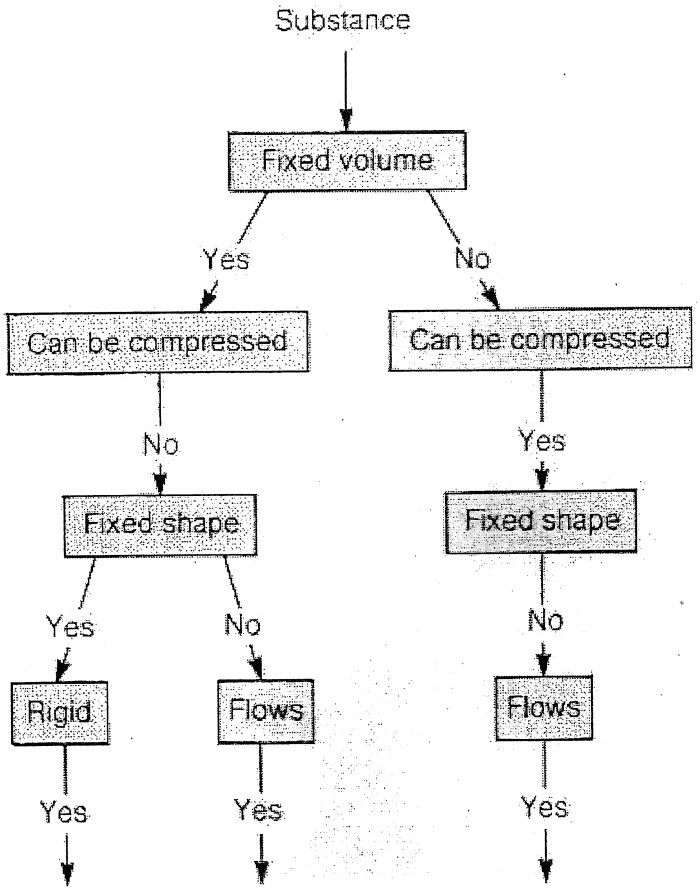
Due: _____

Parent /carer signature: _____

Home reading: Science Essentials 8 Chapter 2.1, 2.2
Homework: Complete the activity below
Reminder: Photosynthesis Investigation is due Friday Wk 8

Solids, liquids, gases

1. Complete the bottom line of the chart below to show the classification of solids, liquids and gases by their properties.



2. Decide if these materials are solids, liquids or gases. Write S, L or G in the space next to each.

Water	____	Rubber	____	Petrol	____
Salt	____	Wood	____	Milk	____
Air	____	Steel	____	Natural gas	____

2.4 Changes of state

Science understanding

FOUNDATION	STANDARD	ADVANCED

The three main states of matter—solid, liquid and gas are shown in Figure 2.4.1.

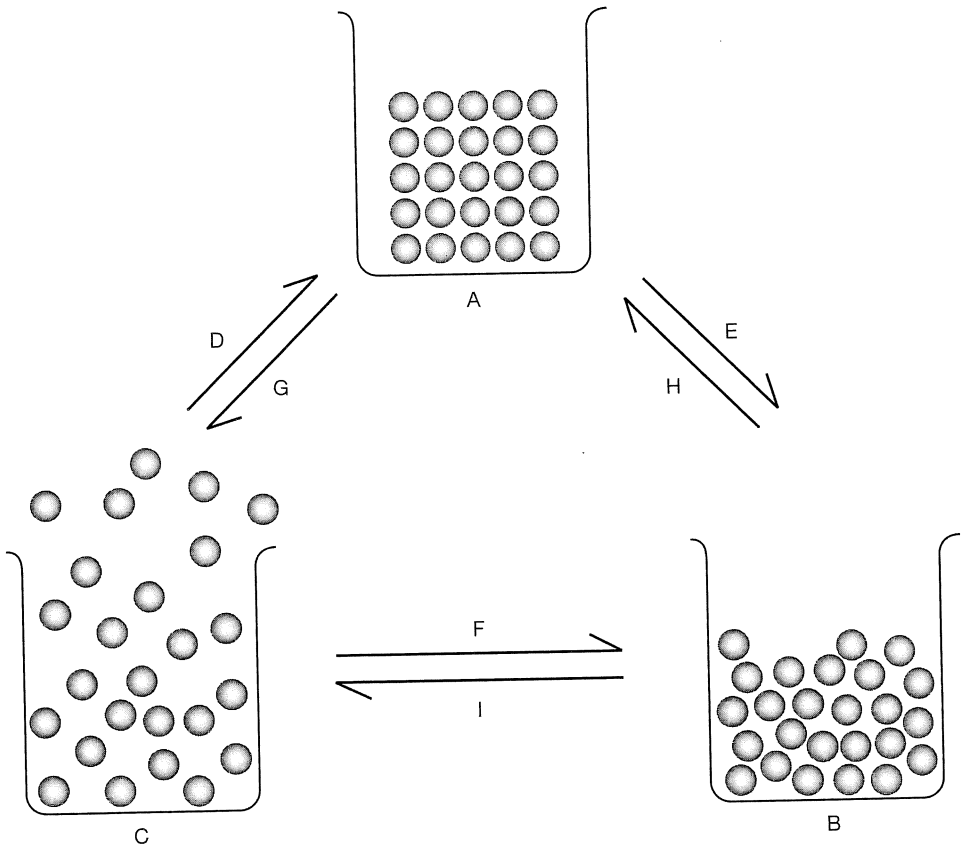


Figure 2.4.1

1 Identify which diagram (A, B or C) best represents a:

(a) solid _____ **(b)** liquid _____ **(c)** gas _____

2 Identify which arrows (D, E, F, G, H or I) represent the following changes of state:

(a) melting _____ **(d)** condensation _____

(b) freezing _____ **(e)** sublimation _____

(c) evaporation _____ **(f)** deposition _____

(3) Explain why the melting point of a substance is the same as its freezing point.

_____

_____